Comparative Analysis of Machine Learning Models for Binary Classification in Metabolomics and Eludication of Key Metabolites

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Abstract

Goal of the project: Comparing performance of four machine learning models in binary classification on a metabolomics data set based on the study by ***Mendez et al.*** (***2019***) and eludicating metabolites most important to the prediction.
Main results: Models performed similarly to the original study, metabolites found predictivly important are shown to be biologically discriptive.
Personal key learnings: Handling model hyperparameter optimization and feature extraction.
Estimated working hours: 10
Project evaluation: 3
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Introduction

Advancements in bioinformatics and data science have significantly contributed to biomedicine, particularly by enabling the analysis of metabolites and their roles in disease mechanisms and cellular functions (***Ali et al., 2024***). However, as these fields advance, it becomes increasingly important to evaluate the performance and reliability of established methods and machine learning models used in biomedical analysis (***Patel and Sharma, 2024***). In the comparative analysis of ***Mendez et al.*** (***2019***) predictive performance of eight different machine learning models was tested and compared on different metabolomics data, to identify the most effective method in metabolomics research.
Here, this comparative study is replicated on four of the eight used machine learning models and additionally elucidates the most predictive metabolites involved in classification.

Goal

The main goal of this project is to recreate the study of binary classification machine learning models on metabolomics data, as presented in ***Mendez et al.*** (***2019***). By consequently comparing the results, the reproducibility of such study can be shown or refuted. Additionally, the most important features of each model are determined in order to elucidate metabolites' potential biological significance for the original study of the dataset (***Ganna et al., 2014***).

Data and Preprocessing

For this project, the dataset MTBLS90 was chosen and obtained via the GitHub repository by ***Mendez*** (***2024***). By downloading the data from the comparative study, it was ensured that the data was already cleaned and was using the Tidy Data framework. It contains data from the ***Ganna et al.*** (***2014***) study, containing plasma samples of 968 participants at age 70 (rows) with 189 metabolite entries (columns) each. The metabolites were elucidated using liquid chromatography-mass spectrometry. The binary classes are female (class 0, 483 samples) and male (class 1, 485 samples). Data was loaded in two tables, where the main data table contained metabolite data in columns to be trained and tested on, and the peak data contained detailed meta information on the columns (metabolites) of the data table.

After import, the main data table is split into features (x) and labels (y) and train and test data following the 2:1 split scheme of the ***Mendez et al.*** (***2019***) paper. Similar to the preprocessing done in the original study of ***Mendez et al.*** (***2019***), all feature data was rescaled using logarithmic transformation and a StandardScaler from the sci-kit learn package (***Pedregosa et al., 2018***). An imputation and binary filtering were not needed, as the data already contains two classes and no missing values.

Methods

Packages used in the study were: sci-kit learn (***Pedregosa et al., 2018***), numpy (***Harris et al., 2020***), (***Wes McKinney, 2010***) and matplotlib (***Hunter, 2007***) for subsequent visualization for this report. Additionally, the use of the cimcb package was refrained from, and the models were reproduced without it.

The four machine learning models chosen were Partial Least Squares Regression, Random Forests and two Support Vector Machines, both with linear and radial bias function kernels. Additionally, for all models, a hyperparameter optimization was carried out where the parameter intervals and values were set similar to the values of the original optimization (***Mendez, 2024***).

PLSR

Partial Least Squares Regression is a method that reduces the data to a smaller number of components in order to highlight variables most important for explaining the relationship between input and output data. According to the studies, PLS evaluates the importance of a variable using Variable Importance in Projection scores (***Chen et al., 2013***). It is set to be optimized in the number of components using grid search, as it has a lesser number of candidates.

Random Forests

Random Forests is a method that constructs multiple decision trees from the data and combines their results to predict the output. According to studies, feature importance is usually measured using the Gini coefficient (mean decrease in impurity). This metric calculates the total decrease in node impurity (***Chen et al., 2013***). The hyperparameter optimization includes the maximal tree depth and minimum number of samples to be present in a leaf node. Here, random search is used to speed up the optimization, as this parameter grid is more extensive.

SVM

Support Vector Machine is an algorithm that finds a hyperplane to distinctly separate the data, where different kernels change how the data is transformed to ensure separation. According to ***Mahadevan et al.*** (***2008***), SVMs measure their feature importance from the absolute values of the model's coefficients, larger absolute coefficient values indicate greater importance. Whereas RBF uses a less straightforward interpretation of feature importance, concisely, it removes features

with the least ranking criterion. For both linear and radial bias function kernels, only the regularization parameter c was optimized using grid search.

Evaluation

After model hyperparameter optimization and training, AUC scores for train and test data were computed using pre-defined sci-kit learn functions.

Following that, for each model, features important to the prediction were extracted. For PLSR, feature importance was extracted by calculating the VIP score with a custom function. For Random Forests and linear SVM, these could be extracted directly using feature_inportances for RF and using coef_ attribute for the linear SVM. Lastly, for SVM with radial bias function kernel feature importances were determined with permutation_importance, which evaluates performance over shuffled feature data, where a bigger drop in performance suggests higher feature importance.

After extraction, values were matched with metabolite names from the peak data table using the metabolite codes in the column names in order to elucidate metabolites important to class distinction.

Results & Discussion

After concluding hyperparameter optimization and model training, following results are achieved: overall, the four chosen models have almost identical AUC_{test} scores (AUC_{test}=0.83) and similar ROC-curves (***Figure 1***). For each model, both AUC scores are close to the reported AUC values in Table 2 from ***Mendez et al.*** (***2019***), with AUC_{train} being slightly more variable and higher than the original values. However, all replicated values, except for RF AUC_{train}, fall into the confidence intervals of the original study.

The variations in the scores can presumably be traced back to different hyperoptimization methods and evaluations of the scores, as here no cimcb package and bootstrapping were used. Overall, the models produce results corresponding to the original study.

Table 1. AUC scores from replicated models and original models in ***Mendez et al.*** (***2019***).

Model	Replicated Values	Original Values
PLSR		
train	0.89	0.88
test	0.83	0.83
RF		
train	0.99	0.91
test	0.83	0.82
SVN - linear		
train	0.88	0.86
test	0.83	0.84
SVN - rbf		
train	0.90	0.89
test	0.83	0.82

In terms of metabolites, the most important features extracted from the test data for all four models were the same, but in different order and expression values. The crucial metabolites given by the models were creatine (M10), Ceramide phosphoethanolamine/Sphingomyelin (M141), and Phosphatidylcholine (M142) (***Table 2***).

First, creatine has been shown to differ by gender, with women having lower creatine levels than

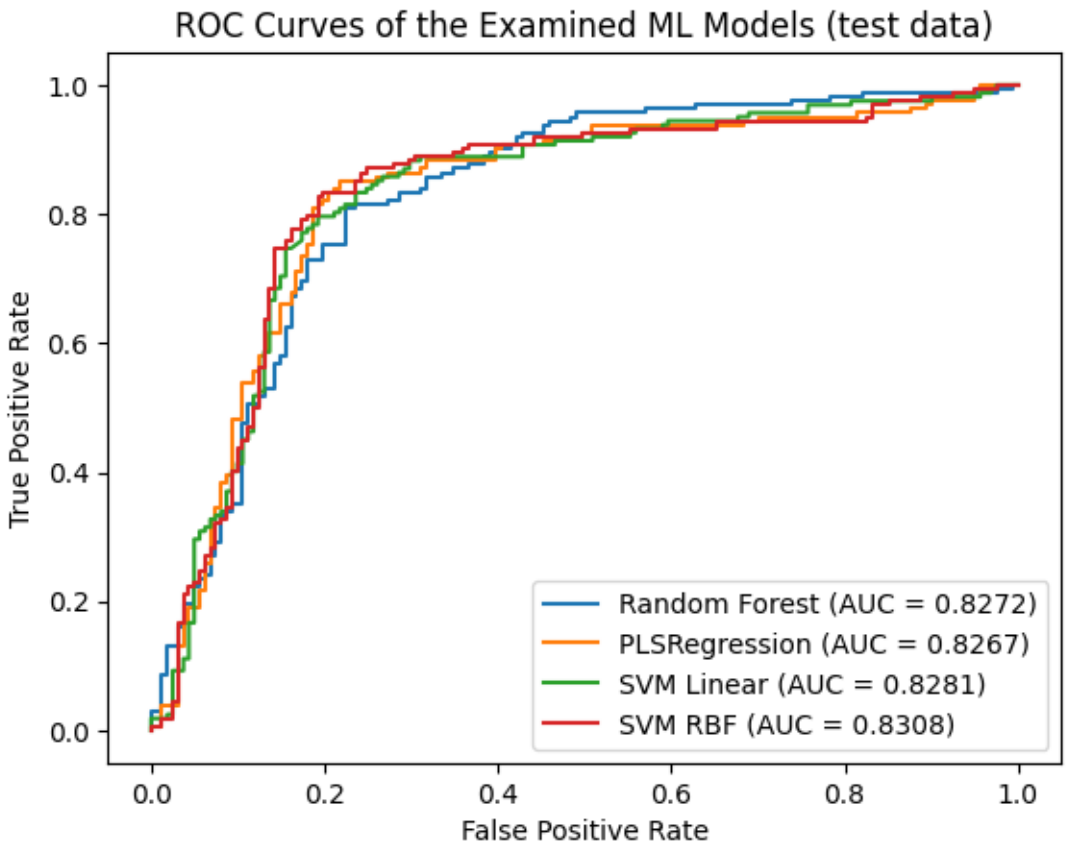


Figure 1. Examined ROC curves for test data of four models considered for evaluation

Table 2. Metabolite Importance in Prediction by Different Models.

Model	Importance	Metabolite	Code
PLSR			
	2.361	Ceramide phosphoethanolamine(35:2)|Sphingomyelin(32:2)	M141
	2.339	Phosphatidylcholine(28:2)	M142
	2.328	Creatine	M10
RF			
	0.059	Ceramide phosphoethanolamine(35:2)|Sphingomyelin(32:2)	M141
	0.058	Phosphatidylcholine(28:2)	M142
	0.036	Creatine	M10
SVN - linear			
	0.106	Creatine	M10
	0.092	Ceramide phosphoethanolamine(35:2)|Sphingomyelin(32:2)	M141
	0.091	Phosphatidylcholine(28:2)	M142
SVN - RBF			
	0.023	Creatine	M10
	0.012	Ceramide phosphoethanolamine(35:2)|Sphingomyelin(32:2)	M141
	0.011	Phosphatidylcholine(28:2)	M142

men, which can be attributed to muscle mass differences (***Gutiérrez-Hellín et al., 2025***). Ceramide phosphoethanolamine and Sphingomyelin have also been shown to be differential between men and women, especially sphingomyelin having higher levels in women, even with age(***Tabassum et al., 2023***). Lastly, phosphatidylcholine levels have also been found to be higher in women than in men, however, the difference in levels can be reversed or even decreased with age (***Tabassum et al., 2023***). Overall, the metabolites that were identified as the ones with the most prediction power on the binary gender classification were also found to relate with gender.

Contributions

1. PLSR and RF models, part of data pre-processing and feature extraction, were done by Ksenia Krasikova.
2. Both SVM models, part of the data preparation and feature extraction, were performed by Payal Priyadarshini.

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